

LendFi

Smart Contract Security Audit

Audit Reference	LFI-2026-01
Commit Hash	1374d96
Audit Date	July 2026
Protocol	https://lendfi-protocol.vercel.app/
Repository	github.com/adityachotaliya9299-jpg/DeFi-Lending-Protocol
Auditor	Aditya Chotaliya
Solidity Version	0.8.24
Network	Ethereum Sepolia Testnet

Critical	High	Medium	Low	Info	Total
0	9	12	10	6	37

1. Executive Summary

LendFi is a production-grade, full-stack decentralised lending protocol built from the ground up, inspired by Aave v3 in architecture. The protocol enables users to deposit crypto assets to earn yield, borrow against collateral at variable or fixed (stable) rates, use flash loans, participate in on-chain governance, build leveraged loop positions, trade interest rate swaps, access structured yield products via tranches, and use NFTs as collateral.

The codebase spans 45 in-scope source files across 20 functional directories, accumulates 670 passing tests across 39 test suites, and has 27 contracts deployed and verified on Ethereum Sepolia testnet. The protocol was developed iteratively over 6 phases from core lending mechanics through derivatives and production hardening.

Severity	Count	Notable
High	9	Supply cap DoS, E-Mode HF bypass, NFT free liquidation
Medium	12	Stable utilization, governance vote snapshot, phantom yield
Low	10	L2 oracle deprecation, dust debt, share inflation
Info	6	Style and documentation notes
TOTAL	37	

The protocol demonstrates strong engineering foundations: comprehensive test coverage, defence-in-depth architecture, and well-documented interfaces. The majority of findings stem from feature integration work in Phases 3-5 where new components were wired together under time pressure. The three most critical issues — supply cap applied to borrow/repay (H-01/02), E-Mode health factor precision mismatch (H-03), and NFT liquidation missing debt collection (H-07) — are straightforward to fix and should be prioritised before any mainnet deployment.

2. Scope

Commit Hash: 1374d96 Branch: main Date Locked: July 1, 2026

core/ (6 files) — HIGHEST PRIORITY

File	Lines	Role
LendingPool.sol	1197 lines	Core: deposit, borrow, repay, withdraw, liquidate
CollateralManager.sol	199 lines	LTV, liquidation threshold, asset config
LiquidationEngine.sol	170 lines	External liquidation helper
FlashLoanProvider.sol	161 lines	Flash loan base contract
BadDebtSocialisation.sol	148 lines	Index reduction for residual debt
CreditDelegation.sol	251 lines	Credit delegation with expiry

oracle/ (3 files)

File	Lines	Role
PriceOracle.sol	248 lines	Chainlink price feed wrapper
OracleAggregator.sol	253 lines	N-source weighted median oracle
OracleWithTWAP.sol	284 lines	TWAP-enhanced oracle

governance/ (3 files)

File	Lines	Role
Governance.sol	140 lines	Protocol parameter governance
GovernanceTimelock.sol	251 lines	Timelocked execution
RiskGovernance.sol	217 lines	Token-weighted risk param voting

tokens/ (3 files)

File	Lines	Role
LendingToken.sol	280 lines	ERC-20 IToken receipt
VariableDebtToken.sol	243 lines	Non-transferable variable debt
StableDebtToken.sol	226 lines	Non-transferable stable debt

Phase 4-6 contracts (30 additional files)

File	Lines	Role
CrossChainMessenger.sol	134 lines	LayerZero-compatible messenger

YieldVault.sol	202 lines	ERC-4626 IToken wrapper
LoopStrategy.sol	251 lines	Leveraged loop position builder
IRSwap.sol	220 lines	Interest rate swap
TrancheVault.sol	265 lines	Senior/Junior tranche structure
NFTCollateralManager.sol	284 lines	ERC-721 collateral
SecurityHardening.sol	218 lines	Circuit breakers + commit-reveal
... + 23 more files		math/, modes/, interfaces/, treasury/, points/, revenue/, liquidation/

Explicitly out of scope: contracts/test/, contracts/script/, contracts/src/mocks/, contracts/lib/ (third-party dependencies).

3. Methodology

Manual Code Review	Line-by-line review of all 45 in-scope contracts against threat model. Every external/public function reviewed for input validation, state mutation ordering, reentrancy exposure, access control, and arithmetic edge cases.
Business Logic Review	Protocol invariants derived from specification and checked against implementation. Key invariants: liquidityIndex/borrowIndex monotonic, getUserDebt reads _scaledBorrows, health factor consistent across modes.
State Machine Analysis	All valid user position transitions mapped (EMPTY → DEPOSITED → BORROWED → REPAID/LIQUIDATED → EMPTY) and every edge case examined.
Access Control Review	Every role enumerated, every onlyRole/onlyOwner modifier verified against intended caller. Centralization risks documented in privilege map.
Oracle Review	Staleness checks, negative price guards, decimal normalisation, and weighted median manipulation resistance all verified.
Flash Loan Review	Reentrancy during callback (nonReentrant guards), repayment enforcement, fee accounting, and available liquidity checks.
Economic Review	Bad debt scenarios, close factor adequacy, liquidation bonus math, interest approximation divergence, TrancheVault bad-debt accounting.
Governance Review	Timelock predecessor correctness, quorum validation, parameter bounds, vote weight integrity.
Cross-Chain Review	Nonce replay protection, trusted remote validation, message ordering, state divergence across chains.
NFT Collateral Review	Floor price oracle centralisation, atomicity constraints, ERC-721 transfer hooks.
Invariant and Fuzz Test Review	670 passing tests reviewed for coverage gaps; fuzz seeds reviewed for edge-case exposure.
Known Vulnerability Pattern Check	Each contract checked against known DeFi patterns: read-only reentrancy, oracle front-running, ERC-4626 donation attacks, governance takeover, integer overflow, precision loss.

Tool	Purpose
Foundry (forge test)	Test execution, fuzz tests, invariant tests
forge coverage --ir-minimum	Branch coverage analysis
Slither	Automated vulnerability detection (static analysis)

Cast	On-chain state queries against Sepolia deployment
Manual Review	Primary review method — line-by-line code inspection

4. Privileged Roles

All privileged roles are currently held by a single deployer EOA (0x72F668Aca488E6d5Aa847f3636aEb0B95413DEF7). On testnet this is acceptable. For mainnet, a Gnosis Safe multisig should hold POOL_ADMIN_ROLE, GUARDIAN_ROLE, and ADMIN_ROLE on all contracts.

Role	Contract(s)	Risk if Abused
POOL_ADMIN_ROLE	LendingPool	Add malicious asset, redirect treasury
GUARDIAN_ROLE	LendingPool	Halt new borrows/deposits indefinitely
ADMIN_ROLE	All Phase 3-6 contracts	Replace oracle feeds, set infinite points rate
ORACLE_ROLE	NFTCollateralManager	Set floor price to 0 or 10x → mass liquidation
POOL_ROLE	PointsAccounting, BadDebtSocialisation	Functional role — held by LendingPool (not EOA)
TREASURY_ROLE	RevenueDistributor	Fund epoch 0 → drain contract
Owner	PriceOracle, InterestRateModel, Governance	Replace Chainlink feeds, set 0% borrow rate
Proposer/Executor	GovernanceTimelock	Schedule/execute parameter changes after delay

5. Findings

[H-01] Supply Cap Validation Applied to Non-Deposit Operations				High
Contract: LendingPool.sol	Function: borrow() / repayWithPermit() / withdraw()	PoC: Yes	Status: Open	
Description	The supply cap check — intended to limit total deposits — is incorrectly called inside borrow(), repayWithPermit(), and withdraw(). Borrow and repay operations do not increase total deposits, yet the check compares current supply + amount against the cap.			
Impact	When total deposits approach the supply cap, all borrow and repay operations revert. Users cannot borrow when the pool has headroom and — critically — cannot repay existing debt. Denial-of-service that locks users into positions they cannot exit.			
Recommendation	Remove checkSupplyCap() from borrow(), repayWithPermit(), and withdraw(). The supply cap should only be enforced in deposit() and depositWithPermit().			

[H-02] Supply Cap Validation Blocks Fund Withdrawals				High
Contract: LendingPool.sol	Function: withdraw()	PoC: Yes	Status: Open	
Description	withdraw() calls checkSupplyCap(asset, currentSupply, withdrawAmount). Since withdrawals decrease total deposits, applying a deposit cap to a withdrawal is semantically inverted. At maximum cap utilization, the check always triggers.			
Impact	Users cannot withdraw their own funds when total deposits are at or near the supply cap. Direct fund-locking vulnerability.			
Recommendation	Remove checkSupplyCap() from withdraw() entirely.			

[H-03] E-Mode Health Factor Uses RAY Instead of WAD — Liquidation Permanently Disabled				High
Contract: LendingPool.sol	Function: _getAccountTotals() E-Mode path	PoC: Yes	Status: Open	
Description	In the E-Mode code path, the health factor is calculated in RAY (1e27) precision: $\text{healthFactor} = (\text{adjustedColl} * \text{RAY}) / \text{totalDebtUsd}$. The comparison constant <code>HEALTH_FACTOR_OK = 1e18 (WAD)</code> . A health factor of 0.5 in E-Mode returns 5e26, which is vastly larger than 1e18. The check always passes.			

Impact	All E-Mode positions are permanently immune to liquidation. Borrowers can maintain arbitrarily undercollateralised E-Mode positions indefinitely, accumulating unrecoverable bad debt.
Recommendation	Change $(adjustedColl * RAY)$ to $(adjustedColl * 1e18)$ in the E-Mode health factor calculation.

[H-04] Stable Debt `_accrueInterest()` Is a No-Op — Interest Lost on Multi-Mint

High

Contract: StableDebtToken.sol	Function: <code>_accrueInterest()</code> / <code>mint()</code>	PoC: Yes	Status: Open
Description	<code>_accrueInterest()</code> only updates <code>_lastAccrualTime</code> without committing accrued interest to <code>_principals</code> . When <code>mint()</code> is called on an existing position, the prior accrued interest is permanently lost. The second mint resets the clock with no interest checkpoint.		
Impact	Protocol systematically undercharges stable-rate borrowers who make multiple borrows. Revenue loss proportional to $(\text{inter-mint time} * \text{original principal} * \text{rate})$.		
Recommendation	Add: <code>_principals[user] += (principal * rate * dt) / RAY</code> ; <code>_ownTotalSupply += interest</code> ; inside <code>_accrueInterest()</code> before resetting <code>_lastAccrualTime</code> .		

[H-05] Zero-Supply Quorum Bypass — Any Token Holder Can Execute Governance

High

Contract: RiskGovernance.sol	Function: <code>execute()</code>	PoC: Yes	Status: Open
Description	Quorum is calculated at execution time: $\text{quorumRequired} = (_getTotalSupply() * \text{MIN_QUORUM_BPS}) / \text{BPS_TOTAL}$. <code>_getTotalSupply()</code> silently returns 0 on failure. The governance token is currently set to USDC. Any USDC holder can propose and execute risk parameter changes with minimal token balance.		
Impact	Governance can be hijacked by any USDC holder. Once <code>_applyParam()</code> is wired to <code>CollateralManager</code> , an attacker can set WETH LTV to 99% and drain the protocol.		
Recommendation	Deploy dedicated ERC20Votes token. Snapshot <code>totalSupply</code> at proposal creation. Revert if <code>totalSupply == 0</code> .		

[H-06] Senior Concentration Cap Skipped on First Deposit

High

Contract: TrancheVault.sol	Function: <code>depositSenior()</code>	PoC: Yes	Status: Open
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Description	Both the 80% senior cap and the junior buffer requirement are gated on <code>currentTotalTVL > 0</code> . The first <code>depositSenior()</code> bypasses both checks, allowing 100% senior TVL with zero junior first-loss buffer.
Impact	The protocol's primary risk mitigation (junior absorbs bad debt first) can be violated on genesis. Senior depositors face full bad-debt exposure with no junior protection.
Recommendation	Remove <code>currentTotalTVL > 0</code> guard. Always require <code>juniorTVL > 0</code> before accepting senior deposits.

[H-07] liquidate() Never Collects Debt Repayment From Liquidator**High**

Contract: NFTCollateralManager.sol	Function: liquidate()	PoC: Yes	Status: Open
Description	liquidate() deletes the position and transfers the NFT to the liquidator but never calls <code>IERC20(borrowToken).safeTransferFrom(msg.sender, ...)</code> to collect debt. The comment reads 'pull from msg.sender in production' — this was never implemented.		
Impact	Any caller can liquidate any undercollateralised NFT position and receive the NFT for free. Borrowers lose their NFT collateral without their debt being repaid.		
Recommendation	Add <code>IERC20(borrowToken).safeTransferFrom(msg.sender, address(this), debt)</code> before deleting the position and transferring the NFT.		

[H-08] 1:1 Price Assumption Breaks All Mixed-Asset Leveraged Positions**High**

Contract: LoopStrategy.sol	Function: openPosition()	PoC: N/A	Status: Open
Description	<code>toBorrow = (currentCollateral * ltvBps) / BPS_TOTAL</code> calculates borrow amount in raw collateral token units, assuming 1:1 exchange rate. For WETH (1e18) collateral and USDC borrow, this attempts to borrow 7e17 USDC (\$700 billion).		
Impact	<code>openPosition()</code> is non-functional for any realistic cross-asset leveraged position. The borrow call will always revert due to insufficient pool liquidity.		
Recommendation	Integrate oracle price conversion: convert collateral value to USD, then convert USD borrow amount to borrowAsset token units using the oracle.		

[H-09] Duplicate _hasBorrow Block — Latent Double-Push Bug**High**

Contract: LendingPool.sol	Function: borrow()	PoC: N/A	Status: Open
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Description	The <code>_hasBorrow</code> tracking block (push to <code>_userBorrows</code> , set <code>_hasBorrow = true</code>) appears twice consecutively in <code>borrow()</code> . The first block executes; the second is dead code because <code>_hasBorrow</code> is already true. Copy-paste residue from Phase 3.1 refactor.
Impact	Currently non-exploitable. If a future refactor moves or removes the first block, the second would execute on an already-tracked position, pushing a duplicate entry to <code>_userBorrows</code> that corrupts health factor iteration.
Recommendation	Remove the second duplicate <code>_hasBorrow</code> block entirely.

5.2 Medium Severity Findings

ID	Title	Contract	Description
M-02	Stable Debt Excluded from Utilization	<i>LendingPool.sol</i>	Interest rates systematically too low when stable borrows exist. Add <code>stableDebt</code> to <code>totalBorrow</code> in interest rate calculation.
M-03	Isolation Debt Ceiling Never Decrement	<i>LendingPool.sol</i>	After liquidation, isolation ceiling never freed. Permanently blocks future borrows on isolated collateral.
M-04	OracleAggregator Median Returns Lower Bound	<i>OracleAggregator.sol</i>	With equal-weight feeds, <code>halfWeight = totalWeight/2</code> always resolves to lowest price. Change to <code>(totalWeight/2)+1</code> .
M-05	Vote Weight Not Snapshot	<i>RiskGovernance.sol</i>	Same tokens can vote multiple times by transferring between addresses during voting window. Use ERC20Votes snapshots.
M-06	No Parameter Bounds in <code>propose()</code>	<i>RiskGovernance.sol</i>	LTV=10000 (100%) can be proposed. Add max bounds per parameter type in <code>propose()</code> .
M-07	<code>distributeYield()</code> Creates Phantom TVL	<i>TrancheVault.sol</i>	TVL incremented without transferring actual tokens. Add <code>safeTransferFrom</code> before incrementing TVL.
M-08	<code>commitLiquidation()</code> Overwriteable	<i>SecurityHardening.sol</i>	Commit-reveal front-runnable via overwrite. Add <code>require(commits[hash].liquidator == address(0))</code> .
M-09	<code>sendMessage()</code> Permissionless	<i>CrossChainMessenger.sol</i>	Anyone can emit cross-chain messages. Add <code>onlyRole(PPOOL_ROLE)</code> .
M-10	Hardcoded Liquidation Buffer	<i>NFTCollateralManager.sol</i>	<code>ltvBps+500</code> hardcoded for all NFT collections. Add configurable <code>liquidationThresholdBps</code> per collection.
M-11	<code>closePosition()</code> Stale Debt Snapshot	<i>LoopStrategy.sol</i>	Pulls <code>pos.totalDebt</code> (principal only); accrued interest orphaned in pool. Query actual debt before pulling.

M-12	Rate Overwrite on Second Stable Mint	<i>StableDebtToken.sol</i>	New rate overwrites old rate globally. Use weighted average rate for multi-borrow users.
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5.3 Low & Informational Findings

ID	Contract	Description
L-01	LendingPool.sol	Redundant second vToken burn in repayWithPermit() — no-op but misleading
L-02	LendingPool.sol	Isolation mode check runs after health factor — wasted gas on failed borrows
L-03	StableDebtToken.sol	Dust principal not cleared after rounding — micro-positions accumulate
L-04	PriceOracle.sol	answeredInRound deprecated on L2 — would brick oracle on Arbitrum/Optimism
L-05	OracleAggregator.sol	Missing round completeness check in _tryGetPrice() (present in PriceOracle)
L-06	GovernanceTimelock.sol	No zero-address check on target in schedule()
L-07	RiskGovernance.sol	_applyParam() silently swallows failures — governance appears to succeed
L-08	CrossChainMessenger.sol	ENDPOINT_ROLE defined but never enforced on lzReceive()
L-09	TrancheVault.sol	Classic 1-wei share inflation attack on first deposit
L-10	SecurityHardening.sol	checkPriceDeviation() emits event (misleading for view callers)
I-01	CollateralManager.sol	ltv=0 not documented — collateral-only intent unclear
I-02	LendingPool.sol	Stable rate hardcoded as 1e15 (development placeholder)
I-03	PriceOracle.sol	Heartbeat silently clamped — should revert to prevent misconfiguration
I-04	PriceOracle.sol	Precision analysis on getValueInUsd() — no issue found
I-05	OracleAggregator.sol	Import statement after inline interface definition (style violation)
I-06	GovernanceTimelock.sol	updateDelay() event ordering confirmed correct — no issue

6. Remediation Review

The following remediation status reflects fixes applied during Phase G of the audit. Each High finding was addressed and re-verified against its PoC exploit.

ID	Finding	Fix Applied	Status	PoC After Fix
H-01	Supply cap on borrow/repay	Remove checkSupplyCap() from borrow(), repay() and repayWithPerf() correctly	Fixed	PoC now reverts
H-02	Supply cap on withdrawal	Same as H-01	Fixed	Covered by H-01 PoC
H-03	E-Mode HF RAY vs WAD	Change RAY to 1e18 in E-Mode HF calculation	Fixed	PoC now shows correct liquidation
H-04	StableDebt accrueInterest no-op	Add interest commit to _principals in _accrueInterest()	Fixed	PoC confirms interest now captured
H-05	Governance quorum bypass	Deploy dedicated gov token; snapshot supply proposal	Pending	New gov token needed (deferred to Phase 7)
H-06	TrancheVault senior cap bypass	Remove currentTotalTVL > 0 guard from depositSenior()	Fixed	PoC now reverts on first senior deposit
H-07	NFT liquidation free loot	Add safeTransferFrom(msg.sender, debtLiquidator, tokenId)	Fixed	PoC now reverts without USDC approval
H-08	LoopStrategy 1:1 price	Added oracle price conversion — TODO: Add oracle integration — deferred to oracle wiring	Pending	
H-09	Duplicate _hasBorrow block	Remove second duplicate block from borrow()	Fixed	Dead code removed
M-07	TrancheVault phantom yield	Add underlying.safeTransferFrom in distributeYield()	Fixed	TVL now matches actual balance
M-08	commitLiquidation overwrite	Add require(commits[hash].liquidator == 0)	Fixed	Front-running now blocked

7. Test Coverage Review

LendFi has one of the most comprehensive test suites reviewed in a Solidity audit of this scope. 670 tests across 39 suites with zero failures provides strong baseline confidence. Notable coverage gaps identified:

Gap	Affected Contract	Severity Impact
E-Mode health factor path never tested with RAY precision checks	LendingPool.sol	H-03 not caught by tests
Multi-mint stable debt interest accrual — no test verified interest	StableDebtToken.sol	H-04 not caught by tests
TrancheVault first deposit tested without junior buffer enforcement	TrancheVault.sol	H-06 not caught by tests
NFT liquidate() called in test but USDC balance/approval never checked	VerifiableCollateralManager.sol	H-07 not caught by tests
RiskGovernance quorum with zero totalSupply not tested	RiskGovernance.sol	H-05 not caught by tests
LoopStrategy openPosition with mismatched decimals not tested	LoopStrategy.sol	H-08 not caught by tests

8. Security Assessment

Code Quality	8.8/10	Well-structured, consistent naming, comprehensive NatSpec. Minor: copy-paste residue in Phase 3.1 refactor.
Documentation	9.0/10	Excellent inline documentation, well-specified interfaces, thorough dev comments throughout.
Test Coverage	9.2/10	670 passing tests, 39 suites, invariant + fuzz tests. Gaps primarily in cross-component integration scenarios.
Architecture	8.5/10	Clean separation of concerns, extensible design. Debt token dual-tracking introduced accidental complexity.
Access Control	8.0/10	Roles correctly defined and enforced. High centralisation risk with single EOA holding all roles (testnet acceptable).
Math & Precision	8.5/10	WadRayMath library sound. Precision mismatch in E-Mode HF (H-03) is the main exception.
Oracle Security	8.5/10	Staleness, negativity, and round completeness checks present. Weighted median algorithm has minor lower-bound bias.
Overall	8.6/10	Production-ready architecture with fixable issues concentrated in Phase 3-5 integration. Recommended: address High findings before mainnet.

9. Conclusion

LendFi represents a technically ambitious full-stack DeFi lending protocol built across 6 development phases, accumulating 670 tests, 27 deployed contracts, and a comprehensive frontend. The codebase demonstrates a mature understanding of DeFi primitives, with well-structured contract interactions, comprehensive interface definitions, and extensive documentation.

The 9 High findings identified are concentrated in three areas: supply cap validation applied to the wrong operations (H-01/02), a precision unit mismatch in E-Mode health factor calculation (H-03), and incomplete implementations in Phase 5 additions (H-06, H-07, H-08). All of these have clear, contained fixes. The StableDebtToken interest accrual no-op (H-04) and RiskGovernance quorum bypass (H-05) require slightly more architectural thought but are equally solvable.

The 6 PoC exploits provided with this report all pass against the pre-fix codebase and confirm the reality of the vulnerabilities. After applying the recommended fixes in Phase G, 7 of 9 High findings are fully resolved and 2 (H-05, H-08) are partially resolved pending architectural decisions (dedicated governance token, oracle integration in LoopStrategy).

Recommendation: Apply all High and Medium fixes before mainnet deployment. Transfer all privileged roles to a Gnosis Safe multisig. Deploy a dedicated ERC20Votes governance token before activating RiskGovernance on mainnet. The protocol is not recommended for mainnet deployment with the current USDC-as-governance-token configuration.

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